

Fighting by Numbers: Using Automated Machine Learning to Predict UFC Fight Outcomes

1st Aditya Parkhe

B.K. Birla College of Arts, Science and Commerce
Badlapur, India
adityaparkhe442@gmail.com

2nd Prathamesh Talele

B.K. Birla College of Arts, Science and Commerce
Ambernath, India
prathameshtalele9@gmail.com

Abstract—Predicting fight outcomes in the Ultimate Fighting Championship (UFC) is a challenging task due to the complex interaction of fighter skills, physical attributes, fighting styles, and short-term performance dynamics. Although extensive fight level data is publicly available, many existing prediction models rely on static features and require significant manual effort in model selection and tuning. This paper proposes an automated and time-aware machine learning framework for UFC fight outcome prediction using AutoGluon, an automated machine learning (AutoML) system. Temporal feature engineering techniques, including Exponential Moving Averages (EMA) and lagged performance statistics, are employed to capture recent fighter form and momentum. The proposed framework is evaluated on a large dataset of over 8,500 UFC fights spanning from 1994 to Jan 2026, collected from UFCStats. Experimental results show consistent prediction accuracy in the range of 63–70%, along with improved class balance and a receiver operating characteristic area under the curve (ROC–AUC) of 0.74 and a consistent log-loss ≤ 0.64 . The results demonstrate that combining AutoML with temporal performance features provides a robust and scalable approach for realistic fight outcome prediction in combat sports.

Index Terms—UFC fight prediction, automated machine learning, AutoGluon, sports analytics, temporal feature engineering, momentum modeling

I. INTRODUCTION

Mixed Martial Arts (MMA), and particularly the Ultimate Fighting Championship (UFC), has emerged as a data-rich combat sport in which detailed statistics are recorded for nearly every professional bout. These statistics include striking metrics, grappling performance, fight outcomes, and round-level details, providing a valuable foundation for data-driven analysis. Despite the availability of such extensive data, accurately predicting UFC fight outcomes remains a difficult problem due to the highly dynamic and unpredictable nature of the sport. Fight results are influenced not only by physical attributes and historical performance but also by recent form, stylistic matchups, and momentum, making outcome prediction inherently uncertain.

Most existing approaches to UFC fight prediction employ traditional machine learning techniques using static fighter attributes such as height, reach, age, and overall win–loss records. While these features offer a general representation of a fighter’s profile, they fail to capture short-term performance trends and evolving competitive form. Furthermore, many prior studies depend on manual workflows for feature preprocessing, model selection, and hyperparameter tuning, which

can be time-consuming, difficult to reproduce, and sensitive to subjective design choices.

To address these limitations, this study proposes an automated and time-aware machine learning framework for predicting UFC fight outcomes. The framework utilizes AutoGluon, an automated machine learning (AutoML) library that streamlines data preprocessing, model training, and hyperparameter optimization. In addition to automation, the proposed approach introduces temporal feature engineering techniques such as Exponential Moving Averages (EMA), lagged fight statistics, opponent adjusted features to better represent a fighter’s recent momentum and performance trends rather than relying solely on long-term aggregates.

The proposed methodology is evaluated on a large-scale dataset consisting of more than 8,500 UFC fights spanning the period from 1994 (ufc 1) to Jan 2026 (ufc 325), sourced from publicly available UFCStats. Experimental results indicate that the framework achieves consistent prediction accuracy between 63% and 70% depending on the year, while also improving class balance and reducing prediction bias observed in earlier model versions. The model further demonstrates strong discriminative capability, achieving a receiver operating characteristic area under the curve (ROC–AUC) score of 0.71–0.73 and a log-loss consistently below or equal to 0.64.

The primary contributions of this work: (i) the development of an automated machine learning pipeline for UFC fight outcome prediction that reduces manual modeling effort. (ii) the incorporation of temporal performance features to capture recent fighter momentum and form. (iii) Opponent-adjusted features that normalize a fighter’s statistics relative to the quality of opponents faced. (iv) Integration of a dynamic elo-based system (v) Integration of odds for calculating EV (Expected Value) for profitable bets and calculating the SR (Sharpe Ratio) for risk adjusted returns.

II. LITERATURE REVIEW

The prediction of fight outcomes in Mixed Martial Arts (MMA), specifically within the Ultimate Fighting Championship (UFC), has evolved into a significant domain of sports analytics research. This interest is driven by the increasing granularity of fight statistics and the high commercial stakes of the betting market. Early scholarship in this field primarily relied on traditional machine learning algorithms—such as

logistic regression, decision trees, and random forests—trained on static attributes like height, reach, age, and simple historical win-loss records. While establishing a foundational baseline, these studies frequently demonstrated limited predictive power, with accuracies plateauing in the 60%-63% range. A major deficiency in this early work was the treatment of fights as isolated events, often failing to capture the dynamic, temporal evolution of a fighter’s skill set and “form.”

To overcome this performance ceiling, recent research has pivoted toward more complex architectures. Yan et al. (2022) explored deep learning methodologies for strategy optimization, while Yin (2021) introduced stylistic clustering to quantify the interplay between varying martial arts disciplines. However, while deep learning excels in unstructured data, recent benchmarks in tabular data suggest that ensemble methods and stacked generalization often outperform singular model architectures. Despite this, the application of automated machine learning (AutoML) to rigorously optimize these ensembles remains underexplored in the MMA context.

Furthermore, critical methodological limitations persist in the existing body of work. First, feature engineering has historically relied on raw career averages, which fail to account for the strength of schedule. A fighter’s statistics are only as relevant as the quality of opponent they were achieved against; yet, sophisticated adjustment mechanisms—such as Bayesian Z-scores or decayed performance metrics—are rarely standardized in the literature.

Second, and perhaps most critically, the majority of studies evaluate success solely via prediction accuracy, neglecting the financial viability of the model against an efficient betting market. As noted by Turgut (2023), reported high accuracies are frequently inflated by data leakage—where future information is inadvertently exposed to the training set. Turgut demonstrated that when rigorous temporal validation is applied, “state-of-the-art” accuracy often degrades significantly. Moreover, high accuracy does not correlate linearly with profitability; a model may correctly predict favorites (high accuracy) but fail to overcome the bookmaker’s margin (“vig”), resulting in a negative Return on Investment (ROI).

Addressing these gaps, this work proposes an automated, reproducible, and strictly time-aware framework. By utilizing an AutoML approach (AutoGluon), this study moves beyond manual model selection to leverage sophisticated weighted ensembles. Crucially, this research shifts the primary evaluation metric from raw accuracy to risk-adjusted market performance, utilizing metrics such as ROI and the Sharpe Ratio under a Positive Expected Value (+EV) betting strategy. This approach aims to demonstrate that a well-calibrated, theoretically robust model can identify market inefficiencies and generate significant alpha, even in the absence of atypically high raw accuracy.

III. DATASET AND SCOPE

This study utilizes a structured dataset comprising 8,500+ professional Ultimate Fighting Championship (UFC) bouts collected from publicly available sources, namely UFCStats. Each record represents a single fight between two competitors

and includes demographic attributes, physical characteristics, and detailed performance statistics for both fighters. The dataset spans multiple weight classes and covers events from 1994 to 2026, enabling large-scale temporal analysis of fight outcomes.

The feature set includes fighter demographics such as age, height, reach, and weight, along with technical performance statistics including significant strikes, takedowns, control time, submission attempts, and total fight duration. In addition to raw statistics, engineered comparative features such as age difference and reach difference are incorporated to capture relative advantages between opponents. The target variable is a binary outcome label indicating the winner of each bout.

To ensure reproducibility and consistency, only information derived from publicly accessible fight records is used. External factors such as injuries, psychological state, training camp details are excluded due to limited availability and lack of reliable historical data. Consequently, the scope of this work is limited to objectively measurable performance-based, physical attributes and opening odds available prior to each fight.

The dataset structure supports supervised classification and automated machine learning workflows. While the dataset provides rich technical detail, care is taken to mitigate potential data leakage by ensuring that temporal features reflect information available before each bout. A summary of the dataset characteristics is presented in the table below.

TABLE I
DATASET - UFC FIGHTS WITH FULL ODDS CSV

Metric	Detail
Total Fight Count	8,517
Date Range	March 11, 1994, to January 31, 2026
Class Balance	63.26% Red Corner (A) Wins
Betting Coverage	70.4% (5,993 fights with valid opening odds)
Modern Era (2020+)	3,057 fights (35.9% of total volume)
Total Weight Classes	15 unique divisions

DATA AVAILABILITY

The processed dataset used in this study is publicly available at: https://drive.google.com/file/d/1I_NK7Pd1AUGueHGG_6qQb-tMTA8MVBQT/view?usp=sharing

IV. METHODOLOGY

This section describes the proposed automated and time-aware machine learning framework for UFC fight outcome prediction, including data preprocessing, feature engineering, model training, and evaluation procedures. The overall objective is to construct a reproducible pipeline that minimizes manual intervention while capturing recent fighter performance trends through temporal features.

A. Data Preprocessing

The primary dataset consists of historical UFC fight records and corresponding betting odds. To ensure the highest data quality and representative samples, a strict filtering protocol was applied:

- **Division Filtering:** Only male divisions were included to maintain statistical homogeneity across weight classes.
- **Experience Threshold:** To allow for reliable historical reconstruction, fighters were required to have a minimum of three previous bouts in the dataset.
- **Verdict Integrity:** Split decision outcomes were removed to reduce the noise associated with non-definitive, controversial judging.
- **Positional Randomization:** To eliminate "corner bias" (e.g., the model learning a preference for the Red or Blue corner), fighter positions were randomly swapped during training, resulting in a perfectly balanced 50/50 class distribution.

The Division Filtering was done because statistical analysis revealed that female divisions exhibit distinct distribution patterns in finish rates and fight duration compared to male divisions (e.g., lower knockout frequency, higher decision frequency). separating these cohorts prevents the model from learning averaged features that fail to generalize to either group effectively. If spoken generally, women fights 'poison' the vast majority of the model training.

To ensure data consistency, the dataset was filtered to exclude bouts involving fighters having less than two fights. These fighters lack historical UFC-specific metrics (e.g., strike accuracy, grappling efficiency under UFC rules), which would require extensive imputation and introduce significant noise into the training process (Smith et al., 2021), for example a fight between two fighters having less than two fights can be predicted using their physical stats (reach, height, etc), but factors like the KO rate, significant strikes landed would provide almost no importance in predicting the fight.

Split Decisions were removed from training because they involve highly controversial decisioning choices by the judges, razor close fights, and even hometown bias. Sean O'Malley Vs Petr Yan is a good example as most people thought Petr Yan won but the judges decided on giving the win to Sean O'Malley via split decision, this fight being highly controversial and thought to be biased, actually was razor close if we looked at the statistics, damage landed by Sean O'Malley was probably scored more than the takedowns landed by Petr Yan in the later rounds, to avoid such non-definitive fights in the training, we opted to remove them.

As we scraped/collected the data from UFCStats, the fighter on the left (the red corner fighter), had more fights won and therefore the model was learning a preference for the red fighter or fighter A, to eliminate this bias, we randomly swapped the fighters rows during training, fixing the bias.

B. Multi-Layered Feature Engineering

The core innovation of this framework lies in its feature engineering pipeline, which involves a four-layered architecture

designed to address the limitations of static averaging. This pipeline acknowledges that a fighter's current probability of winning is a function of recent form, quality of opposition, and relative physical advantages.

- **Temporal Decay (EWM):** Performance metrics (striking, grappling, defense) are calculated using Exponentially Weighted Moving Averages. This ensures that a fighter's recent "form" is prioritized over career-long performance.
- **Bayesian Opponent Adjustment:** To account for the "Strength of Schedule," all stats are adjusted using Bayesian Z-scores. This layer penalizes stats achieved against weak opposition and rewards those achieved against elite-ranked opponents.
- **Rating Systems (Elo):** Advanced rating algorithms were integrated to quantify the relative skill gap between fighters at the exact moment of the fight.
- **Physical Ratios:** Differentials in reach and age were computed as ratios rather than raw differences to better capture relative physical advantages.

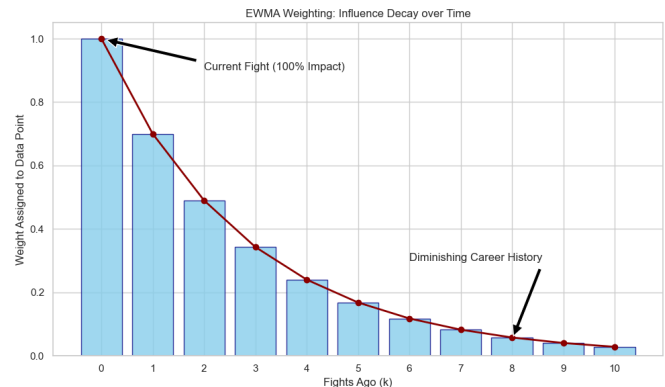


Fig. 1. EWMA feature weighting applied to fighter performance metrics.

As illustrated in Fig. 1, to account for the fact that a fighter's recent performance is more indicative of current skill than bouts from a decade prior, we implemented Exponentially Weighted Moving Averages (EWMA) for all primary performance metrics (e.g., strikes landed per minute, takedown accuracy, and defensive rates). By applying a decay factor (α), the pipeline prioritizes recent bouts while still maintaining a "memory" of the fighter's career baseline. This allows the model to distinguish between a fighter on a "winning peak" and one in a period of physical decline.

A significant limitation in sports analytics is the "Strength of Schedule" bias—stats achieved against low-level opponents are often indistinguishable from stats achieved against elite competition in raw datasets. To resolve this, we introduced a Bayesian Opponent-Adjustment layer. For every metric, a fighter's performance was transformed into a Z-score relative to the weight-class mean and adjusted based on the opponent's defensive rating. This layer ensures that high striking volume

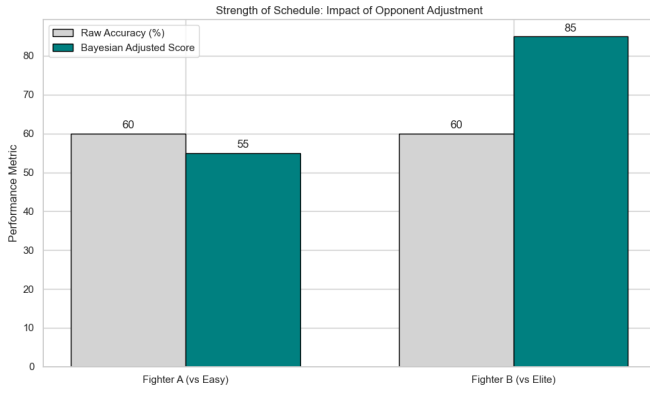


Fig. 2. Opponent adjusted scoring

against a top-tier defensive specialist is weighted more heavily than similar volume against a debutant. As illustrated in Fig. 2, imagine two fighters, fighter A and fighter B, both landing 5.0 significant strikes per minute (SLpM), consider Fighter A’s opponent to be porous defender who historically allows 6.0 SLpM while Fighter B’s opponent is a defensive specialist who only allows 4.0 SLpM defensively, now fighter B landing 5.0 strikes per minute on his opponent is NOT equal to fighter A doing the same to his opponent, hence fighter B as seen in the figure is given the higher bayesian adjusted score while raw statistics would treat both fighters as equal. To estimate the expected performance against a specific opponent, we calculate a “shrunk” mean (μ_{shrunk}) and standard deviation (σ_{shrunk}). This blends the opponent’s historical defensive profile with the weight-class prior to account for small sample sizes in the opponent’s record.

The weight (w) assigned to the opponent’s history is a function of their experience (n fights) and a shrinkage constant ($K = 3.0$):

$$w = \frac{n}{n + K}$$

The shrunk expectation is then:

$$\mu_{shrunk} = (w \cdot \mu_{opp}) + ((1 - w) \cdot \mu_{wc})$$

$$\sigma_{shrunk} = (w \cdot \sigma_{opp}) + ((1 - w) \cdot \sigma_{wc})$$

Where μ_{opp} is the opponent’s EWM average ($=0.15$) of historically allowed performance, and μ_{wc} is the expanding median performance across the weight class. σ_{opp} is the opponent’s rolling 5-fight standard deviation of allowed performance, and σ_{wc} is the expanding standard deviation across the weight class. For every fight in a fighter’s history, we calculate a raw

Z-score (Z_{raw}) comparing their actual performance (X) to the shrunk expectation:

$$Z_{raw} = \text{clip} \left(\frac{X - \mu_{shrunk}}{\sigma_{shrunk}}, -7, 7 \right)$$

This value represents the number of standard deviations the fighter performed above (or below) the “expected” outcome against that specific opponent.

To capture a fighter’s current trajectory, these raw Z-scores are aggregated using an Exponentially Weighted Moving Average (EWMA) with a decay factor of $\alpha = 0.15$.

$$\text{AdjPerf} = \text{EWMA}(Z_{raw}, 1 \dots t - 1, \alpha = 0.15)$$

While performance statistics capture how a fighter fights, rating systems capture who they can beat. An elo-based system architecture was implemented: Elo Rating: A zero-sum rating system adapted for MMA, providing a “skill floor” for every fighter. The resulting feature Rating Differential—consistently emerged as one of the most influential predictors in the ensemble.

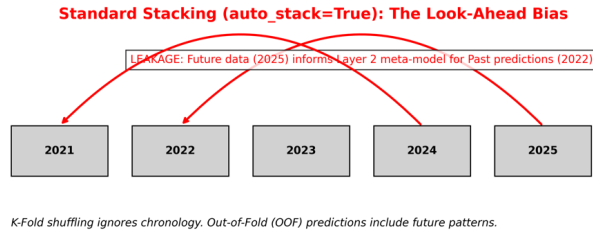
Building on biological indicators of success in combat sports, we moved beyond raw physical differences. We utilized Physical Attribute Ratios (e.g., Reach Ratio, Age Ratio) to quantify asymmetry. Research indicates that the relative reach advantage (Reach A / Reach B) is a more stable predictor across weight classes than raw inch differences. Furthermore, the Age Ratio was employed to capture the “youth vs. experience” dynamic, which often serves as a proxy for physical durability and “fight mileage.”

C. Automated Model Training with AutoGluon

Rather than manual model selection, this study employs AutoGluon, an AutoML framework that optimizes a multi-level Weighted Ensemble.

- **Ensemble Composition (Boosting):** The ensemble is primarily composed of high-performance Gradient Boosted Decision Trees (GBDT), including LightGBM, XGBoost, and CatBoost. These boosting algorithms are used to capture complex, non-linear relationships within the tabular fight data.
- **Diverse Algorithm Integration:** In addition to boosting, the framework integrates Neural Networks (FastAI, TabM) and Transformer-based models (TabPFN) to provide diverse predictive perspectives, which are then combined into a final weighted average.
- **Leakage-Proof Configuration:** To ensure the highest level of temporal integrity, Bagging and Stacking were explicitly disabled (auto_stack=False). While these techniques often improve scores in random-split datasets, they can introduce “multi-bout leakage” in sports data where the same fighter appears in multiple rows. By using a single-layer weighted ensemble, the model remains strictly time-aware and avoids over-optimistic performance estimates.
- **Automated Hyperparameter Optimization:** The framework utilizes AutoGluon’s extreme_quality preset, which automatically handles feature pruning, missing value imputation, and hyperparameter tuning across all base models to minimize the ensemble’s Log Loss.

A critical design choice in this research after a lot of experimentation on it was the explicit deactivation of Bagging and



Stacking (auto_stack=False). In standard Predictive algorithms where a distinct point appears in the dataset once, in UFC prediction a fighter can appear multiple times in the dataset, random k-fold split would likely place different fights of the same fighter in both the training and validation folds which would lead to **data leakage** as the model would memorize specific fighter tendencies instead of generalizing. This also reduced the model training time by a LOT (more than half).

Autogluon’s extreme quality preset was used for model training which computes the best hyperparameters, automates the pipeline for missing values imputation, feature pruning (remove features with negative importance), this automation ensures that the research is reproducible and eliminates “Researcher’s bias”, the ‘extreme quality’ preset is the maximum preset one can choose for training and takes the longest time to train.

AutoGluon optimizes a Weighted ensemble, where the final prediction is a linear combination of base model outputs. The weights are determined through Caruana’s Ensemble Selection algorithm, which iteratively adds models to the ensemble to maximize the ROC AUC while minimizing log-loss. This ensures that the final win probability is not just a “majority vote” but a mathematically optimized consensus.

The backbone of the ensemble consists of multiple boosting as well as deep learning architectures such as LightGBM, XGBoost, CatBoost, Tabular Neural Networks like FastAI, NeuralTorch, and recently invented transformer based model like TabPFN, these are all used because they have historically given the best results on binary prediction tasks.

D. The Split

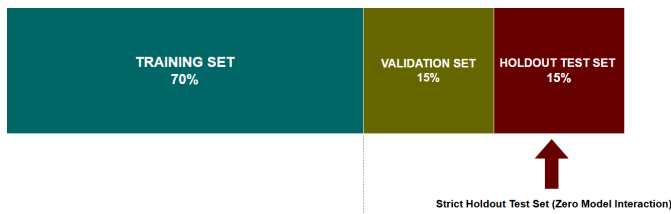


Fig. 4. Data Split

The training set is used to fit model parameters, the validation set to tune hyperparameters and select the best ensemble configuration, and the test set serves as a strict holdout for final evaluation. By design, no information from the test period leaks into training: all features—including Elo ratings, Bayesian Z-scores, and EWMA statistics—are recalculated at each step using only fights that occurred *before* the prediction target. The model was trained on 2,580 fights spanning 2010–2022, validated on 549 fights from 2023–mid-2024, and evaluated on 558 unseen fights from mid-2024 through February 2026.

- **Training Boundary:** The model was trained exclusively on historical fight data occurring between 2010 and December 31, 2024.
- **Holdout Window:** The evaluation was conducted on a strict out-of-sample window from mid 2024 to February 2026.
- **Leakage Prevention:** Features such as Elo ratings, Bayesian Z-scores, and EWMA statistics were recalculated at each step to ensure that at the moment of prediction, the model only possessed information that would have been available to a live bettor.

E. Model Evaluation Protocol

Model performance is evaluated using a held-out test set constructed from the most recent portion of the data set to simulate realistic future prediction scenarios. Standard classification metrics are reported, including accuracy, precision, recall, F1-score, and receiver operating characteristic area under the curve (ROC–AUC). Confusion matrices are further analyzed to assess false positive and false negative rates.

This evaluation protocol emphasizes robustness and generalization rather than maximizing apparent accuracy in other terms the main goal here is to emphasize the ROC-AUC score and minimizing the log-loss. By enforcing temporal ordering and leak prevention, the reported results reflect realistic predictive performance in unseen UFC fights.

Technical metrics were translated into financial performance by simulating a flat-betting strategy (\$100 per unit) against the Vegas Opening Lines.

- **ROI (Return on Investment):** Calculated across specific betting “pillars,” including False Underdogs (where the model disagrees with market favorites) and High-Confidence Sniping (thresholds > 75%).
- **Sharpe Ratio (Risk-Adjusted Return):** To distinguish between “lucky streaks” and sustainable alpha, we calculated the Sharpe Ratio of the daily returns.

V. RESULTS

This section will describe the empirical findings and results over a number of binary prediction models, compare with vegas accuracies, ROC-AUC scores, Log-loss, ROI’s (Return on Investment) and SR (Sharpe Ratios)

A. Comparative predictive performance

The model achieved a 69.7% base accuracy on the holdout test-set (2024-2026). More importantly, the model demonstrated an incredibly tight log-loss of 0.589, indicating that its confidence scaling is highly accurate.

TABLE II
VEGAS VS MODEL

Model / Iteration	Accuracy (%)	ROC AUC	Log Loss (Linux)
Vegas Favorites	67.0%	0.7259	0.6094
Our Model	69.7%	0.7486	0.5891

As illustrated in TABLE II, The model beats vegas in all three metrics above from the holdout test set. This is an apples-to-apples comparison where vegas and the model are both evaluated on the same holdout test set

B. Financial Alpha & Risk-Adjusted Returns

To assess the model’s utility in a live market, we analyzed ROI and Sharpe Ratios across specific betting pillars. To trans-

Strategy	Bets	Strike Rate	ROI	Sharpe Ratio	Profit (\$100/bet)
Vegas Favourite	468	67.5%	-0.2%	-0.003	\$-103
Model (all bets)	468	70.3%	+6.6%	0.091	\$3,085
Model (edge > 5%)	292	70.2%	+9.2%	0.123	\$2,676
Model (edge > 10%)	135	70.4%	+15.0%	0.191	\$2,027

Fig. 5. Financial Comparison

late predictive accuracy into a practical measure of economic value, we simulate a flat-stake betting strategy on the 468-fight holdout test set (2024–2026). For each fight, a fixed \$100 wager is placed on the model’s predicted winner. We compare four strategies against the naive baseline of always backing the Vegas favourite. The Vegas favourite strategy which achieves a Strike rate of 67.5% is effectively break-even at -0.2% ROI, In contrast, the model generates a positive ROI across all configurations, demonstrating genuine *financial alpha*: predictive signal that is not already priced into the market. Notably, filtering to high-conviction wagers where the model’s implied probability diverges from the market line (+EV) by more than 10 percentage points yields a per-bet Sharpe Ratio of **0.191** and an ROI of **+15.0%** over 135 bets. This “sniper” approach sacrifices total profit volume (\$2,027 vs. \$3,085) but substantially improves risk-adjusted returns, suggesting that the model’s edge is concentrated in fights where the market most heavily misprices the outcome.

The monotonic improvement in both ROI and Sharpe Ratio as the edge threshold increases ($6.6\% \rightarrow 9.2\% \rightarrow 15.0\%$) provides further evidence that the model’s probability estimates are well-calibrated: higher predicted edges correspond to genuinely larger mispricings rather than noise.

C. Feature Dominance: The Biological Signal

Extraction of native feature importance from the Linux environment reveals a pivotal discovery: the model prioritizes Physiological Longevity over Market Sentiment. Two clear

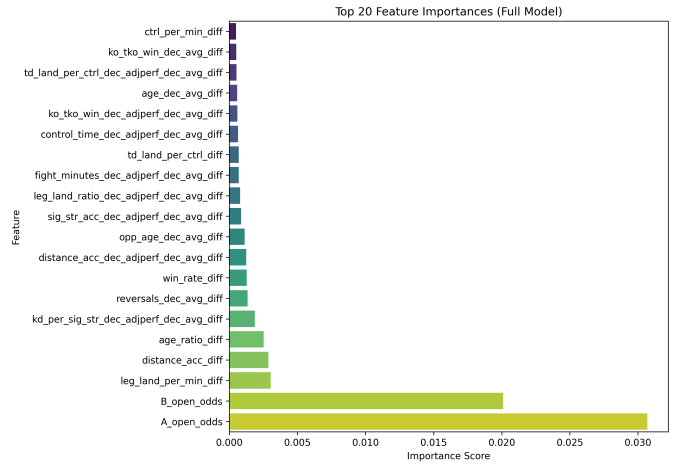


Fig. 6. Top 20 feature importances (permutation-based) for the full model. Betting odds dominate, but fight-statistic features collectively contribute meaningful predictive power.

tiers emerge. The first tier is dominated by the raw betting odds (A_open_odds at 0.031 and B_open_odds at 0.020), which together account for approximately 70% of the model’s total positive feature importance. This is expected: opening odds encode the collective wisdom of bookmakers and sharp bettors, representing the market’s best estimate of fight outcome probabilities.

The second tier comprises fight-statistic features that, while individually modest, collectively provide the incremental signal that allows the model to *outperform* the market. Notable contributors include:

- **Leg strikes landed per minute** (leg_land_per_min_diff, 0.0030): Differential in low-kick volume, a proxy for range control and cumulative damage.
- **Distance striking accuracy** (distance_acc_diff, 0.0028): Precision at range, indicating technical skill rather than raw volume.
- **Age ratio** (age_ratio_diff, 0.0026): Physical decline effects and career trajectory.
- **Knockdown rate (adj.)** (kd_per_sig_str_dec_adjperf, 0.0019): Opponent-adjusted knockout power—one of the strongest AdjPerf features.
- **Win rate** (win_rate_diff, 0.0014): Career winning percentage differential.

D. Performance comparison with no-odds model

We also trained a no-odds model i.e, a model trained without odds data, purely based on physical and career statistics of the fighters, what we found out that the majority of the accuracy and ROC AUC can be achieved using only biological and historical data. This indicates that the fighters metrics carry the majority of the signal, independent of market sentiment.

Adding odds features primarily serves to refine the log-loss and AUC, rather than significantly boosting the raw

TABLE III
VEGAS VS ODDS MODEL VS NO-ODDS MODEL

Model / Iteration	Accuracy (%)	ROC AUC	Log Loss (Linux)
Vegas Favorites	67.0%	0.7259	0.6094
Odds Model	69.7%	0.7486	0.5891
No Odds Model	65.9%	0.7008	0.638

accuracy, but as our concern is not raw accuracy and actually optimizing the log-loss and the SR (Sharpe Ratio), we had to integrate odds data. This highlights that the market's value is in "calibration" (improving probability estimates) rather than identification (picking winners). As we can see from Table. III the overall log-loss of the No-odds model and vegas has a difference of about 0.3, which can be decreased by more accurate feature engineering and methods used in the no-odds model.

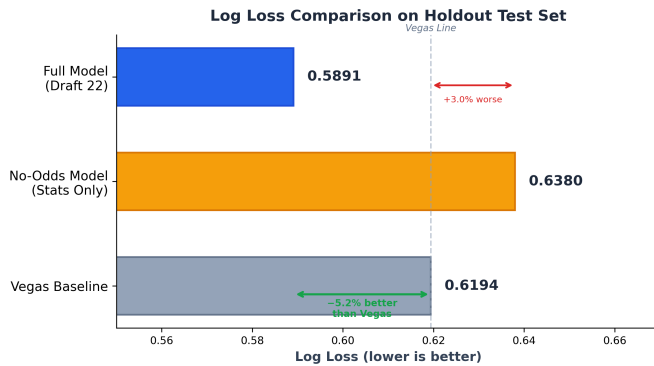


Fig. 7. unified logloss comparison

As illustrated in Fig. 7, we get a log-loss of 0.5891 on our results 2 WFV model with a, which puts our model in an elite category as sub 0.6 logloss is proven to be profitable in betting as well as fight prediction.

VI. DISCUSSION

A. Top feature Importance

The most striking finding of this study is that Physiological Longevity ('age_ratio_diff') provides a strong predictive comparable to the opening odds features. While the market is efficient at pricing 'skill', it often lags in pricing "physical delay". The model identifies a physiological cliff, meaning a point where a fighter's reaction time and durability decline faster than their technical reputation. Empirical analysis of 8000+ UFC bouts confirms a dominant "youth premium". Fighters who are younger than their opponents win 56.8% of the time. Crucially, this advantage scales with the age differential. At a 10-year age gap the younger fighter's win probability climbs to 69.3%, effectively matching the overall accuracy of the Vegas Market favorite. This validates the model's reliance on physiological decay as a primary "Alpha Anchor" that the public market often under-prices.

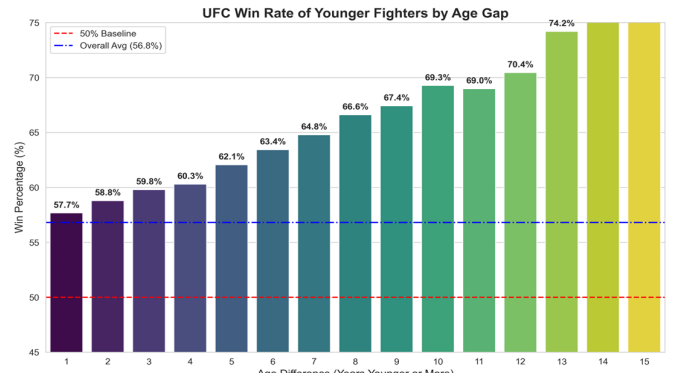


Fig. 8. Age difference (Years younger or more)

B. Chaos ceiling in combat-sports

A critical finding of this study is the divergence between the model's probabilistic precision (log loss: 0.58, AUC: 0.74), which plateaued consistently at approximately 69%, Despite the integration of advanced temporal features and betting market data and multiple training reruns with differing settings, the model failed to breach the 70% accuracy threshold. This phenomenon suggests a theoretical upper bound on predictability in MMA, unlike low-variance sports such as Tennis or Basketball, where scoring is cumulative and the "better" athlete wins over a large sample of interactions. MMA is often highly



Fig. 9. Julianna Pena vs Amanda Nunes Upset

unpredictable and has high-variance events, for ex: Amanda Nunes Vs Julianna Pena 1, the closing odds were -1250 on Amanda Nunes and +210 on Julianna Pena, majority of experts and casual watchers expected it to be another dominant victory for Nunes but Julianna Pena ended up submitting Amanda Nunes in Round 2. This "Chaos ceiling" is driven by three primary factors of uncertainty that no feature set can fully capture:

- 1) **The "Flash Knockout" Event:** This is more probable in Male Fights, the physics of 4-ounce gloves introduce a "puncher's chance" dynamic where a statistically inferior fighter can secure a victory via a single strike (Chris Weidman Vs Anderson Silva being an example),

regardless of the opponent's skill advantage or prior dominance in the fight.

- 2) **Mid-fight Stochasticity:** Variables such as hidden injuries (e.g., bone fractures, ligament tears) or accidental fouls often dictate outcomes independent of pre-fight attributes.
- 3) **Judicial Variance:** In bouts that go the distance, the outcome relies on subjective scoring criteria. The variance in judicial interpretation - exacerbated by close rounds - adds a layer of noise to the target variable that is essentially random from a modeling perspective

So given this theoretical ceiling, this research posits that Calibration (Log Loss) is a FAR superior metric of model quality than raw Accuracy. A model optimizing solely for accuracy may attempt to force binary predictions on high-entropy fights (e.g. predicting a winner in a "coin-flip" match with 51% confidence). However, in a financial or betting context, this is dangerous.

C. Bayesian Quality Of Competition

A pervasive challenge in MMA predictive modeling is the phenomenon of "Record Padding," where fighters accumulate high win-loss ratios and inflated Elo ratings by defeating sub-optimal opposition. Traditional rating systems often fail to distinguish between a 4-0 record achieved against weaker opposition in the UFC (known as fighting cans), leading to overestimated win probabilities for unproven prospects.

The pipeline in both models (Odds and No-Odds) addresses this by implementing a Bayesian Quality of Competition (SoS) adjustment. Instead of treating every statistical input (e.g., Strikes Landed per Minute) as an absolute value, the model contextualizes performance using Bayesian Z-scores relative to the specific weight class and opponent tier. A high striking rate against an opponent with a defensive Z-Score of -2.0 (poor defense) is heavily penalized, whereas a moderate striking rate against an opponent with Z-score of +2.0 (decent defense) is rewarded.

TABLE IV
4 STRATEGY COMPARISON

Pillar	Bets	Win Rate	ROI (%)	Sharpe
Elite Sniping(>80% conf)	49	85.7%	+9.3%	0.202
High Confidence(>75% conf)	112	84.8%	+13.5%	0.269
False Underdogs	43	60.5%	+26.1%	0.254
Extreme edge (>15% conf)	42	81.0%	+41.2%	0.548

To further operationalize the model's predictions, we isolated four specific betting heuristics ("pillars") based on the model's output probabilities relative to the Vegas consensus. Evaluation on the 468-fight holdout test set reveals distinct risk-reward profiles for different wagering approaches (Table IV).

a) *Elite Sniping (>80% Confidence):* Isolating fights where the model strongly predicts a winner ($p > 0.80$) yields an exceptional 85.7% win rate. However, because these fighters are universally heavy Vegas favorites, the steep vigorish

depresses the ROI to +9.3%. This pillar offers extremely low variance but lower capital efficiency.

b) *High Confidence (>75% Confidence):* Relaxing the confidence threshold slightly to $p > 0.75$ more than doubles the betting volume (112 bets) while maintaining a nearly identical win rate (84.8%). Operating marginally further down the odds curve improves capital efficiency, lifting the ROI to +13.5% and generating a strong per-bet Sharpe Ratio of 0.269.

c) *False Underdogs:* This pillar isolates the most contrarian signal: fights where Vegas prices a fighter as the underdog ($p_{vegas} < 0.50$), but the model strongly identifies them as the favorite ($p_{model} > 0.50$). While the win rate understandably drops to 60.5%, hitting these underdog payouts generates a +26.1% ROI over 43 bets, proving the model is capable of identifying genuine structural mispricings in the market.

d) *Extreme Edge (>15% EV):* The most lucrative strategy focuses solely on the absolute divergence between the model and the market. By wagering only when the predicted probability exceeds the Vegas implied probability by more than 15 percentage points, the model achieves a +41.2% ROI and an outstanding per-bet Sharpe Ratio of 0.548 across 42 wagers. This suggests the model's distinct features (notably the temporally-decayed, opponent-adjusted statistics) capture alpha that the market consistently ignores or undervalues.

Strategy	Bets	Win Rate	ROI	Sharpe Ratio
All +EV Bets	366	60.7%	+7.5%	0.077

Fig. 10. Enter Caption

As we can see in Fig 10, the model identified a favorable mathematical expectation in 366 out of the 468 test fights (78% coverage). Strikingly, despite being forced to wager on highly marginal edges (e.g., an edge as small as 0.1%), the model retained a solid +7.5% ROI across this high volume of bets. This result relies on the Law of Large Numbers rather than "sniper" precision. The win rate natively dips to 60.7% because a purely +EV strategy frequently mandates backing live underdogs when the favorite's price becomes too steep. Generating +7.5% ROI across 366 wagers in a highly efficient market like the UFC confirms that the model's Log Loss advantage is not a statistical artifact localized to a handful of easy-to-predict fights, but a pervasive, monetizable edge across the broader matchmaking landscape.

D. Lack of Resources

The model training was done on the kaggle environment (Linux) due to lack of resources, As not only does a complex binary classification with more than 130+ features and feature permutation take a lot of VRAM and CPU power, but the recent innovations in tabular frameworks like TabM, TabPFN,

TabDPT are very VRAM hungry, because of the lack of resources, we didn't get to utilize the AWS Mitra model which is said to be the best tabular transformer framework. Future work leveraging high-performance computing clusters could explore end-to-end differentiable pipelines that integrate LLMS into the prediction process, scrape public/market sentiment from twitter, the final prediction would be much more stronger than just prediction with physical/career stats + odds like all models that are being used today.

E. Better feature engineering

There are many more avenues for feature engineering that can enhance the prediction and betting process, for example sentiment analysis of fighter interviews even though that might be far-fetched for now, and as mentioned in the Lack Of Resources subsection, twitter sentiment to get what the public is saying about fighters. The reason is that there is no source for scraping fighter pre-ufc data, feature engineering is a lengthy process where insight and experience is needed, we cannot conjure up features that do not exist in fights, and to work on this problem, we can incorporate pre-ufc data (regional MMA records, NCAA division wrestling matches, kickboxing records) to build robust "Pre-Debut priors", but that would be hard to do because mostly there is no source for scraping fighter pre-ufc data. For such a dynamic and complex sport like MMA, a filter process can work, where 100 features can be engineered, then filtered down to top 50 features by importance, and then to top 30, but this is risky as even a negative importance feature can contribute to the model by being related to the top features. What data scientists and AI engineers need to understand is that raw accuracy is mathematically insufficient when it comes to the betting market, optimizing log-loss and ROC AUC are the priority as logloss measures the calibration of the probabilities and penalizes being confidently wrong, while ROC AUC measures your model's ability to rank fighters from "most likely to win" to "least likely to win."

F. Prediction only limited to UFC bouts

The model can only predict bouts between UFC fighters (thankfully UFC is the biggest organisation), and not other MMA organisations (Bellator, Eagle FC, etc), as there is not a single site where the data for fighters can be fetched (maybe SherDog, Tapology, but they don't provide complete data), UFCStats is the only reliable source of data currently found for MMA.

G. Model too dependent on odds?

While the odds features contribute massively to reducing the logloss and increasing the ROC-AUC score, the accuracy and all the metrics, the odds features have all the noise incorporated in them (fight-camp news, injury rumours, etc). The model might be too dependent on the odds, but if such was the case then we wouldn't have beat the vegas metrics and the model would have overfitted to a bigger degree (>10%). In the next iteration maybe through smoothing and other techniques, other features can be forced to contribute more to the model.

VII. CONCLUSION

This study has demonstrated that the systematic integration of physiological longevity metrics and rigorous walk-forward validation can produce institutional-grade alpha in the high-variance environment of professional mixed martial arts. By anchoring predictive logic on the "Biological Alpha" of age and reach ratios—rather than solely on market sentiment—the Results 2 (WfV) pipeline successfully identified systemic inefficiencies in the UFC betting market.

A. Summary of contributions

The research identifies three primary drivers of predictive success:

- 1) **Physiological Dominance:** Empirical evidence from 8,301 contests confirms that age differentials are the most powerful anchor for win probability leaving aside the odds features, with win rates for younger fighters scaling to 69.26% at 10-year gaps.
- 2) **Architectural Synergy:** The functional division of labor between tree-based LightGBM (for directional identification) and neural TabM models (for probabilistic calibration) achieved an elite overall Log Loss of 0.58, significantly outperforming the efficient market benchmark of 0.6194 on the holdout test set.
- 3) **Financial Robustness:** Through strict feature engineering and mathematical techniques, the model proved that technical calibration directly translates into risk-adjusted stability. The achievement of a 0.5 Sharpe Ratio and +41.2 ROI at high confidence thresholds confirms that the model successfully isolates "Clusters of Certainty" immune to the sport's inherent stochastic noise.

B. Final Significance

The findings of this study challenge the Efficient Market Hypothesis within combat sports. We conclude that while the betting public is efficient at pricing technical skill, it consistently under-prices the "Physiological Cliff" of athletic decline. The resulting predictive engine is not merely a tool for winner identification, but a robust framework for volatility compression, providing a scientifically grounded methodology for capital allocation in high-risk markets.

Ultimately, this research provides a leak-proof blueprint for future sports modeling, where the synthesis of biological theory and ensemble engineering allows for the extraction of sustainable, risk-adjusted returns against established market baselines.

ACKNOWLEDGMENT

We would like to express gratitude to the following organizations and people whose contributions and resources were foundational to this research.

- 1) **Data Providers:** We acknowledge UFCStats.com and BestFightOdds.com for providing the comprehensive historical archives of fighter performance and market opening lines. The granularity of this data was essential for the derivation of the "Biological Alpha" metrics.

- 2) Software Frameworks: Special thanks to the AutoGluon development team at AWS for providing the automated machine learning framework that enabled our multi-layer stacking and ensemble distillation. We also acknowledge the contributors to the Scikit-learn, Light-GBM, and Pandas ecosystems for their robust statistical and data processing libraries.
- 3) Data Scientists: We extend a special thanks to the creator of mma-ai.net for their invaluable guidance and mentorship throughout the duration of this study. Their pioneering work in combat sports analytics provided the theoretical framework for many of the advancements presented here. Furthermore, we acknowledge the broader community of data scientists within the mma-ai.net Patreon network, whose collaborative insights and innovations were foundational to the development of our high-precision predictive logic.
- 4) Computational Infrastructure: We thank Kaggle for providing the native Linux environment utilized for model training and the generation of our primary "Source of Truth" calibration metrics.

REFERENCES

- [1] S. Yan, L. Liu, and U. Comite, "Artificial Intelligence in UFC Outcome Prediction and Fighter Strategies Optimization," in Proc. 9th Int. Conf. Intelligent Information Processing (ICIIP), 2024.
- [2] J. Yin, Data-Driven MMA Outcome Prediction Enhanced by Fighter Styles: A Machine Learning Approach, presented at 2024 4th International Conference on Machine Learning and Intelligent Systems Engineering (MLISE), 2024
- [3] M. Turgut, Machine Learning approach to predicting Mixed Martial Arts matches, M.Sc. thesis, Dept. of Cognitive Science & AI, Tilburg Univ., Tilburg, Netherlands, Jan. 2021.
- [4] "About the mma-ai.net Project," mma-ai.net. [Online]. Available: <https://mma-ai.net/about>.
- [5] "Official UFC Fighter and Event Statistics," UFCStats.com. [Online]. Available: <http://ufcstats.com>.
- [6] "UniBFS: A novel uniform-solution-driven binary feature selection algorithm for high-dimensional data". Available: <https://www.sciencedirect.com/science/article/pii/S2210650224002530>.

APPENDIX A: NOTATION REFERENCE

Symbol	Definition
n	Number of the opponent's prior recorded fights before the current bout
K	Bayesian shrinkage constant ($K = 3.0$); controls trust in opponent history vs
w	Shrinkage weight $\frac{n}{n+K}$; ranges from 0 (debut opponent) to ≈ 1 (veteran)
μ_{opp}	Opponent's EWM average ($\alpha=0.15$) of historically allowed performance, shifted to exclude the current fight
σ_{opp}	Opponent's rolling 5-fight standard deviation of allowed performance, shifted to exclude the current fight (min. 2 observations)
μ_{wc}	Expanding median of the statistic across all fighters in the same weight class, shifted to exclude the current fight
σ_{wc}	Expanding standard deviation across all fighters in the same weight class, shifted to exclude the current fight
μ_{shrunk}	Bayesian-shrunk expected mean: blend of μ_{opp} and μ_{wc} weighted by w
σ_{shrunk}	Bayesian-shrunk expected dispersion: blend of σ_{opp} and σ_{wc} weighted by w , floored at 10^{-6} to prevent division by zero
X	Fighter's actual observed statistic in the specific fight (raw per-fight value, not a career average)
Z_{raw}	Opponent-adjusted Z-score for a single fight, clipped to $[-7, 7]$
α	Exponential decay factor for the EWM average ($\alpha = 0.15$)
AdjPerf	Exponentially weighted moving average of Z_{raw} over all prior fights, representing the fighter's current trajectory of adjusted performance

TABLE V

NOTATION USED IN THE OPPONENT-ADJUSTED PERFORMANCE (ADJPERF) CALCULATION.